

UNIT VI

Trends in Cloud Computing

By
Dr. C P Ahir
LCO
GP Amravati

The Place of computer technology in our lives...



Mainframe Comp.

many people share
a computer

Use must be well
prepared

"run by experts
behind closed doors"



Personal Comp.

one computer, one
person

direct explicit use

"while it may take you
where you want to go,
it requires
considerable attention
to operate"



Ubiquitous Comp.

many computers share each of
us

Use implicit (automatic)

"each person is continually
interacting with hundreds of
nearby interconnected
computers"

Ubiquitous Computing (UbiCom)

- Found every where.

Example:

- Advertisement of new buildings

Ubiquitous Computing (UbiCom)

- Ubiquitous computing (or "ubicom") is a concept in software engineering and computer science where computing is made to appear anytime and everywhere.
- In contrast to desktop computing, ubiquitous computing can occur using **any device**, in **any location**, and in **any format**.
- A user interacts with the computer, which can exist in many different forms, including laptop computers, tablets and terminals in everyday objects such as a fridge or a pair of glasses.

Ubiquitous Computing (UbiCom)

- The underlying technologies to support ubiquitous computing include
 - ☐ Internet,
 - ☐ advanced middleware,
 - ☐ operating system,
 - ☐ mobile code,
 - ☐ sensors,
 - ☐ microprocessors,
 - ☐ Virtual Reality
 - ☐ new I/O and user interfaces,
 - ☐ networks, mobile protocols,
 - ☐ location and positioning and new materials.
- Enable computer-based services to be made available everywhere (Ubiquitous)

History

Mark Weiser:
the father of ubiquitous computing



- researcher in the Computer Science Lab at Xerox's PARC (Palo Alto Research Center)
- first articulated the idea of ubiquitous computing in 1988
- has called UC "...highest ideal is to make a computer so embedded, so fitting, so natural, that we use it without even thinking about it."

Ubiquitous Computing

- * **Pervasive computing, also called ubiquitous computing**, is the growing trend of embedding computational capability (generally in the form of microprocessors) into everyday objects to make them effectively communicate and perform useful tasks in a way that minimizes the end user's need to interact with computers as computers.
- * Pervasive computing devices are network-connected and constantly available.

Ubiquitous Computing

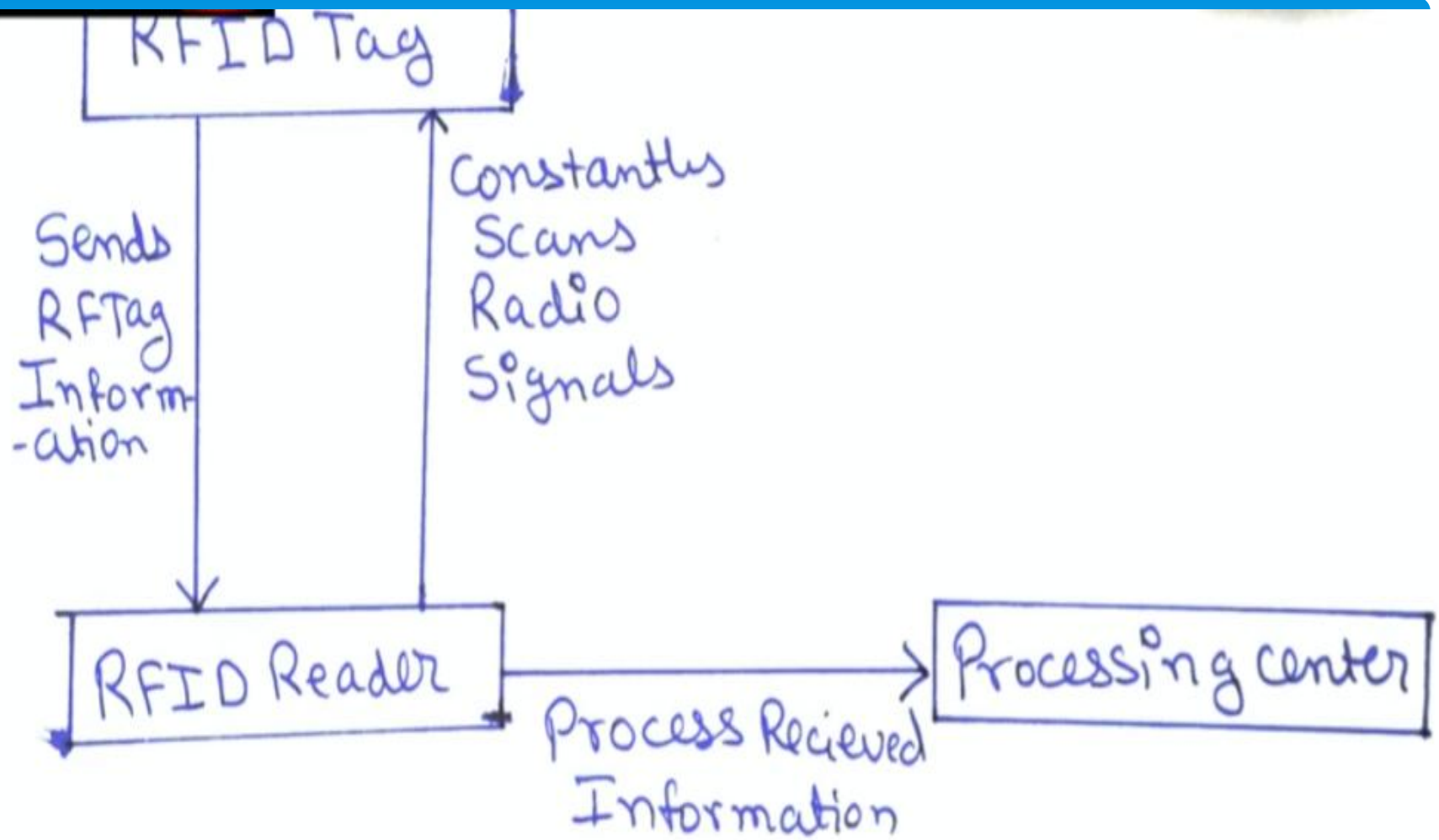
- * An example of pervasive computing is an [Apple Watch](#) that alerts the user to a phone call and allows the call to be completed through the watch.
- * Another example is when a registered user for Audible, Amazon's audio book server, starts his or her book using the Audible app on a smartphone on the train and continues listening to the book through Amazon Echo at home.
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Ubiquitous Computing

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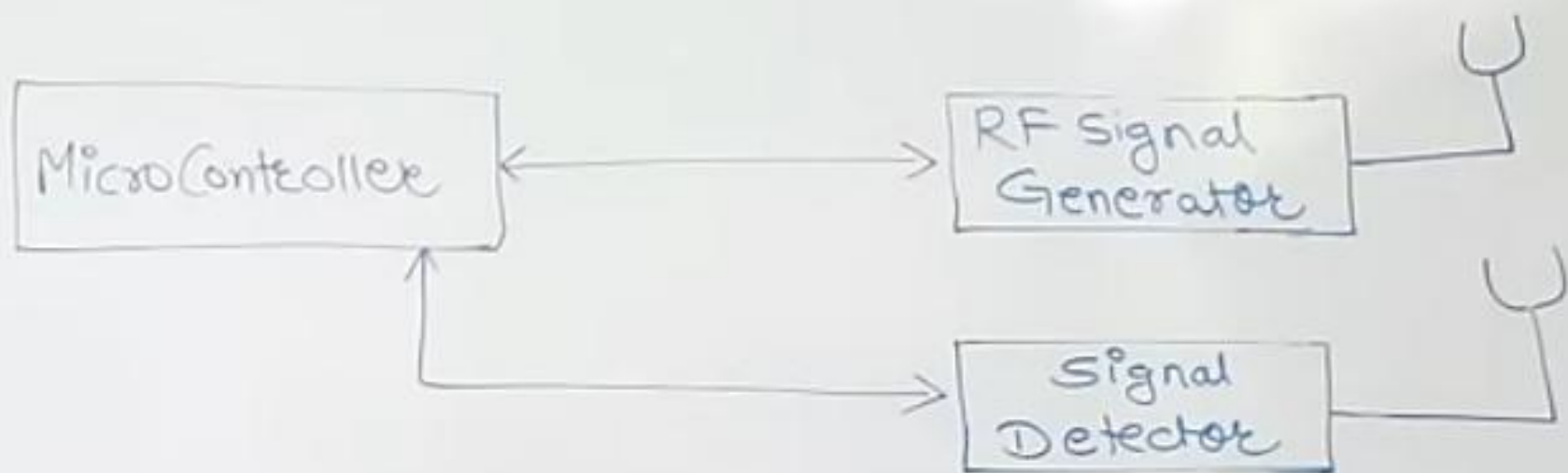
Internet of Things

- * The internet of things, or IoT, is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers ([UIDs](#)) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.
- * An IoT ecosystem consists of web-enabled smart devices that use embedded systems, such as processors, sensors and communication hardware, to collect, send and act on data they acquire from their environments.



RFID Diagram

RFID



RFID (radio frequency identification)

- * RFID is an acronym for “radio-frequency identification” and refers to a technology whereby digital data encoded in RFID tags or smart labels (defined below) are captured by a reader via radio waves.
- * At a simple level, RFID systems consist of three components: an RFID tag or smart label, an RFID reader, and an antenna.
- * RFID tags contain an integrated circuit and an antenna, which are used to transmit data to the RFID reader (also called an interrogator).
- * The reader then converts the radio waves to a more usable form of data.

RFID (radio frequency identification)

- * **RFID TECHNOLOGY IS EMPLOYED IN MANY INDUSTRIES TO PERFORM SUCH TASKS AS:**
- * – Inventory management
- Asset tracking
- Personnel tracking
- Controlling access to restricted areas
- ID Badging
- Supply chain management
- Counterfeit prevention (e.g. in the pharmaceutical industry)

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Zigbee

- * Zigbee is a standards-based wireless technology developed to enable low-cost, low-power wireless machine-to-machine (M2M) and internet of things (IoT) networks.
- * For example, other wireless technologies like WiFi and Bluetooth are striving to go faster, add more features, provide streaming of high definition content and run for hours.
- * In contrast, the Zigbee Technology is designed for low data rates, fit into a tiny stack, controlling simple devices like LEDs or Thermostats and run for years.

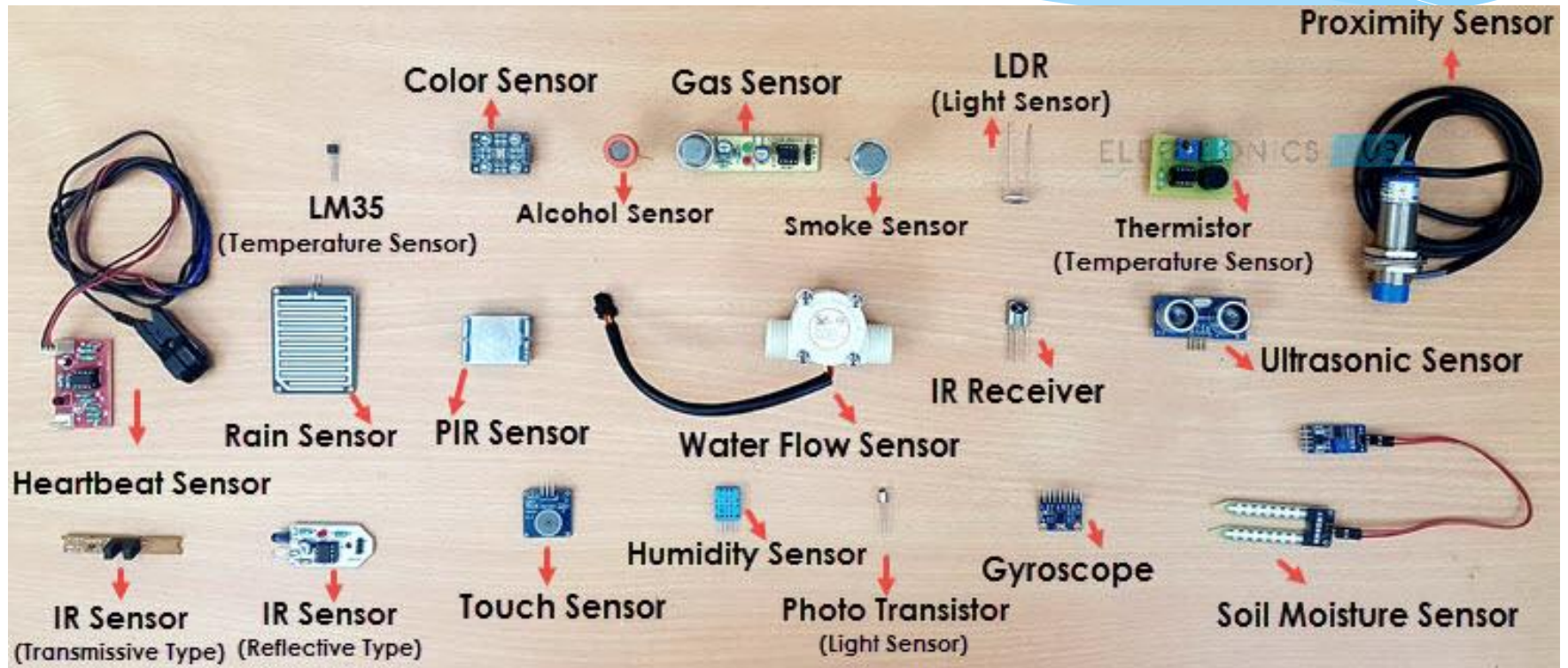
Zigbee Applications

- * Home Automation
 - * Security Systems
 - * Meter Reading Systems
 - * Light Control Systems
 - * HVAC Systems
- * Consumer Electronics
 - * Gaming Consoles
 - * Wireless Mouse
 - * Wireless Remote Controls
- * Industrial Automation
 - * Asset Management
 - * Personnel Tracking
 - * Livestock Tracking
- * Healthcare
- * Hotel Room Access
- * Fire Extinguishers

Sensor Technologies

- * sensors are devices that detect the feature quantity of a measurement object and convert this quantity into a readable signal, which is displayed on an instrument.
- * And sensing technology, simply put, is a technology that uses sensors to acquire information by detecting the physical, chemical, or biological property quantities and convert them into readable signal.

Sensor Technologies



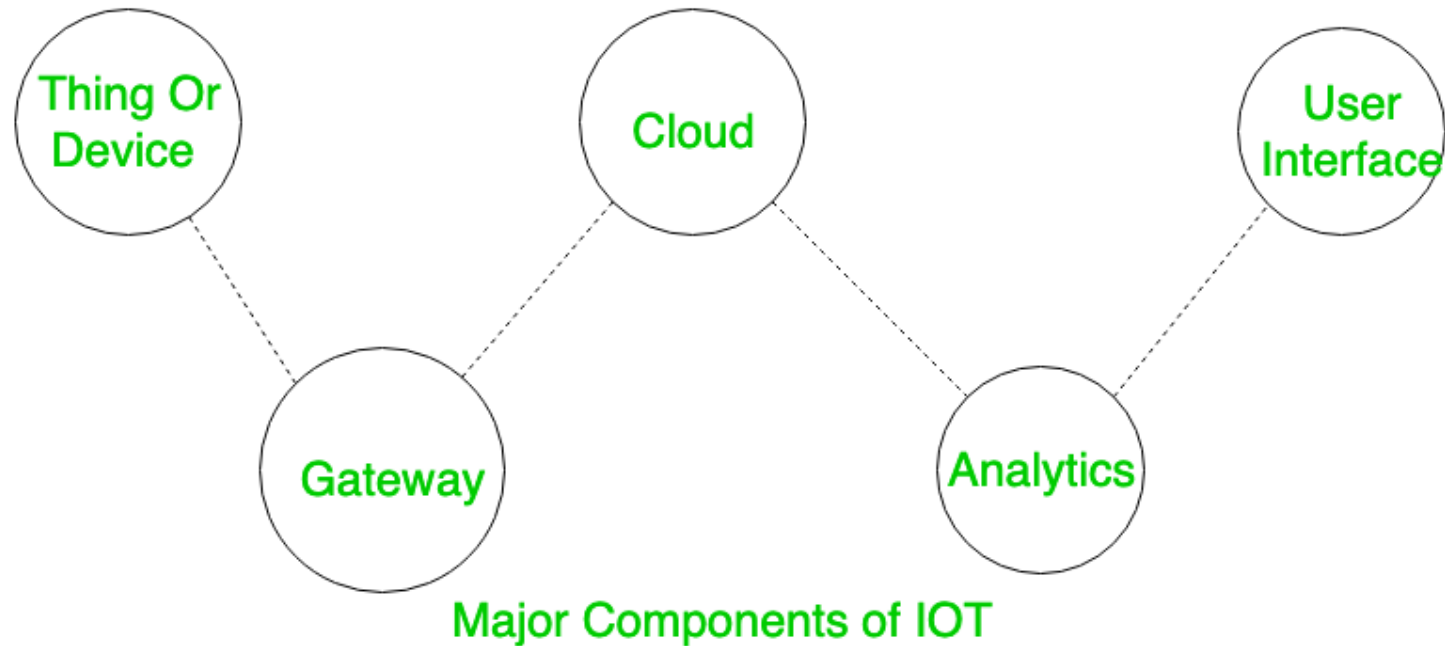
GPS

- * The Global Positioning System (GPS) is a space-based radio-navigation system consisting of a constellation of satellites broadcasting navigation signals and a network of ground stations and satellite control stations used for monitoring and control.
- * Currently 31 GPS satellites orbit the Earth at an altitude of approximately 11,000 miles providing users with accurate information on position, velocity, and time anywhere in the world and in all weather conditions.

Components of IOT and relation with Cloud Computing

- * in simple terms IOT means anything that can be connected to internet and can be controlled / monitored using Internet from our smart devices or PCs.
- * The “things” specified here can be anything from small tracking chips to actual smart cars on road, all these can be categorized as IOT.

Components of IOT and relation with Cloud Computing



Components of IOT and relation with Cloud Computing

- * **Things or Device**

- * These are fitted with sensors and actuators. Sensors collect data from the environment and give to gateway where as actuators performs the action (as directed after processing of data).

- * **Gateway**

- * The sensors give data to Gateway and here some kind of pre-processing of data is even done. It also acts as a level of security for the network and for the transmitted data.

- * **Cloud**

- * The data after being collected is uploaded to cloud. Cloud in simple terms is basically a set of servers connected to internet 24*7.

Components of IOT and relation with Cloud Computing

- * **Analytics**

- * The data after being received in the cloud processing is done . Various algorithms are applied here for proper analysis of data (techniques like Machine Learning etc are even applied).

- * **User Interface**

- * User end application where user can monitor or control the data.

How Does Cloud Computing Help to make IoT better?

- * **1. Cloud computing helps IoT devices by providing a platform for data storage and analysis.**
- * IoT devices are becoming increasingly popular as they offer a convenient way to connect to the internet and collect data.
- * However, these devices often lack the storage and processing power needed to effectively use this data.
- * Cloud computing helps IoT devices by providing a platform for data storage and analysis.
- * This allows users to more easily access and use the data collected by their devices.

How Does Cloud Computing Help to make IoT better?

- * **2. Cloud computing helps IoT devices by providing a platform for data sharing and collaboration.**
- * IoT devices are becoming increasingly popular as they offer a convenient way to connect to the internet and share data.
- * However, these devices can be difficult to manage and keep track of without a cloud-based platform.
- * Cloud computing helps IoT devices by providing a platform for data sharing and collaboration.
- * This allows users to access their data from any device with an internet connection, and makes it easier to manage large numbers of devices.

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How Does Cloud Computing Help to make IoT better?

- * **3. Cloud computing helps IoT devices by providing a platform for data security and privacy.**
- * IoT devices are constantly about their surroundings and sending it back to a central server.
- * This data is then used to improve the performance of the device and make it more efficient.
- * However, this data is also valuable to hackers who could use it to exploit the device or steal sensitive information.
- * Cloud computing helps IoT devices by providing a platform for data security and privacy. This way, only authorized users can access the data and it is more difficult for hackers to obtain.

How Does Cloud Computing Help to make IoT better?

- * **4. Cloud computing helps IoT devices by providing a platform for scalability and flexibility.**
- * Cloud computing helps IoT devices by providing a platform for scalability and flexibility.
- * This allows IoT devices to tap into the vast resources of the cloud and perform complex tasks that would otherwise be impossible.

How Does Cloud Computing Help to make IoT better?

- * **5. Cloud computing helps IoT devices by providing a platform for cost savings.**
- * By using cloud services, IoT devices can offload some of their processing and storage requirements to the cloud, which can reduce their overall costs.
- * In addition, cloud services can provide a way for IoT devices to share data and communicate with each other, which can further reduce the costs of operating these devices.

Cloud Platforms: Amazon EC2 and S3

* **Amazon Elastic Compute Cloud**

- * Amazon EC2 provides scalable computing capacity in the Amazon Web Services (AWS) Cloud.
- * Using Amazon EC2 eliminates your need to invest in hardware up front, so you can develop and deploy applications faster.
- * You can use Amazon EC2 to launch as many or as few virtual servers as you need, configure security and networking, and manage storage.
- * Amazon EC2 enables you to scale up or down to handle changes in requirements or spikes in popularity, reducing your need to forecast traffic.

Features of Amazon EC2

- * Amazon EC2 provides the following features:
 - Virtual computing environments, known as *instances*
 - Preconfigured templates for your instances, known as *Amazon Machine Images (AMIs)*, that package the bits you need for your server (including the operating system and additional software)
 - Various configurations of CPU, memory, storage, and networking capacity for your instances, known as *instance types*
 - Secure login information for your instances using *key pairs* (AWS stores the public key, and you store the private key in a secure place)
 - Storage volumes for temporary data that's deleted when you stop, hibernate, or terminate your instance, known as *instance store volumes*

Features of Amazon EC2

- Persistent storage volumes for your data using Amazon Elastic Block Store (Amazon EBS), known as *Amazon EBS volumes*
- Multiple physical locations for your resources, such as instances and Amazon EBS volumes, known as *Regions* and *Availability Zones*
- A firewall that enables you to specify the protocols, ports, and source IP ranges that can reach your instances using *security groups*
- Static IPv4 addresses for dynamic cloud computing, known as *Elastic IP addresses*
- Metadata, known as *tags*, that you can create and assign to your Amazon EC2 resources
- Virtual networks you can create that are logically isolated from the rest of the AWS Cloud, and that you can optionally connect to your own network, known as *virtual private clouds (VPCs)*

Amazon Simple Storage Service (Amazon S3)

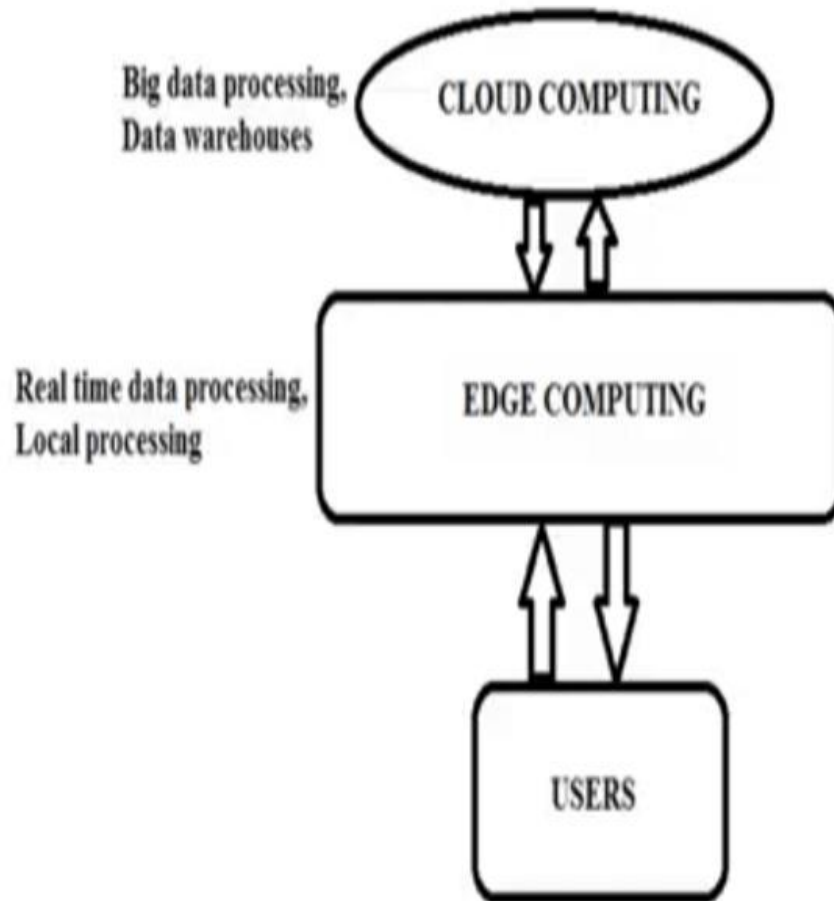
- * It is an object storage service that offers industry-leading scalability, data availability, security, and performance.
- * Customers of all sizes and industries can use Amazon S3 to store and protect any amount of data for a range of use cases, such as data lakes, websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics.
- * Amazon S3 provides management features so that you can optimize, organize, and configure access to your data to meet your specific business, organizational, and compliance requirements.

How Amazon S3 works

- * Amazon S3 is an object storage service that stores data as objects within buckets.
- * An object is a file and any metadata that describes the file. A bucket is a container for objects.
- * To store your data in Amazon S3, you first create a bucket and specify a bucket name and AWS Region.
- * Then, you upload your data to that bucket as objects in Amazon S3. Each object has a key (or key name), which is the unique identifier for the object within the bucket.

How Amazon S3 works

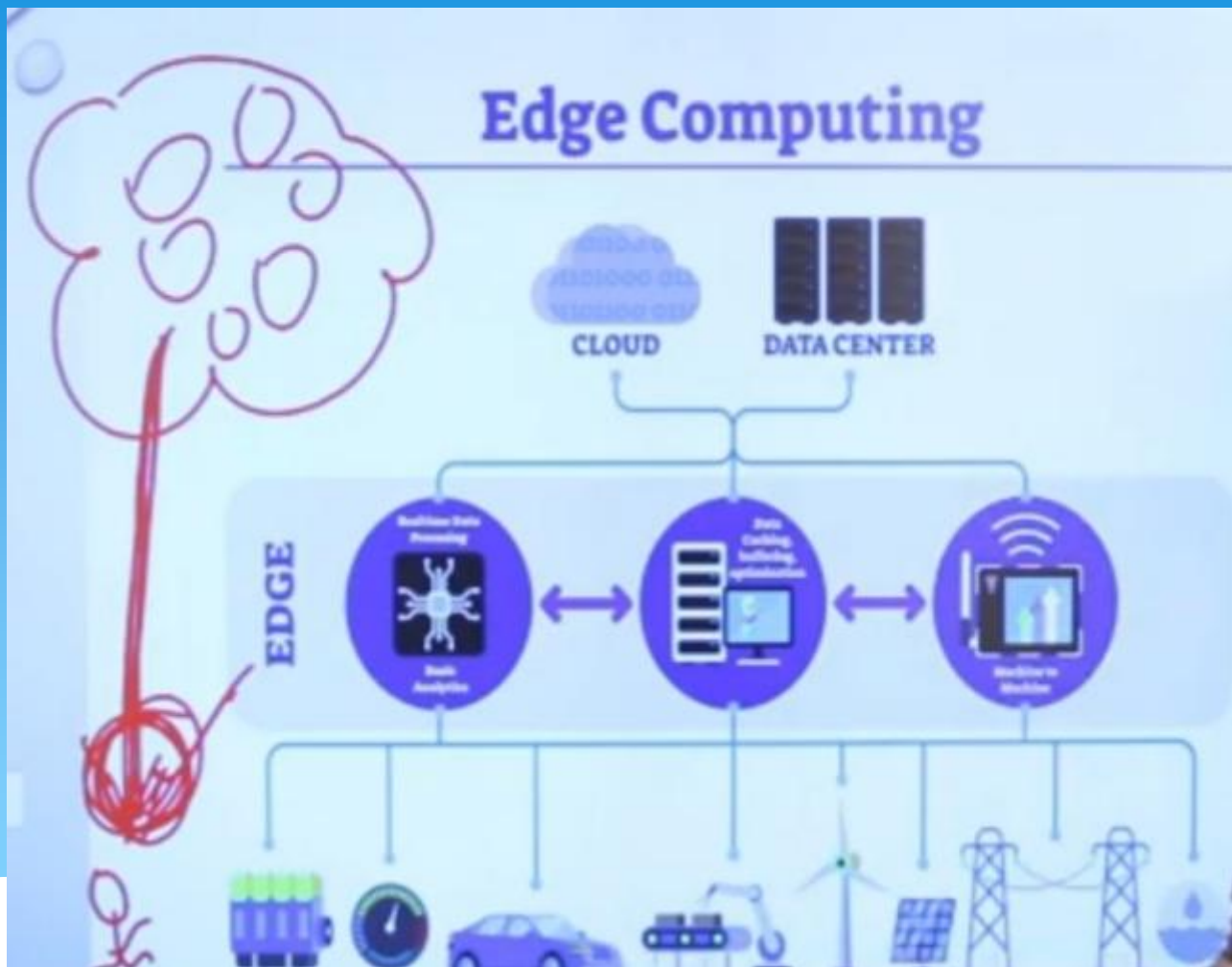
- * S3 provides features that you can configure to support your specific use case.
- * For example, you can use S3 Versioning to keep multiple versions of an object in the same bucket, which allows you to restore objects that are accidentally deleted or overwritten.
- * Buckets and the objects in them are private and can be accessed only if you explicitly grant access permissions.
- * You can use bucket policies, AWS Identity and Access Management (IAM) policies, access control lists (ACLs), and S3 Access Points to manage access.

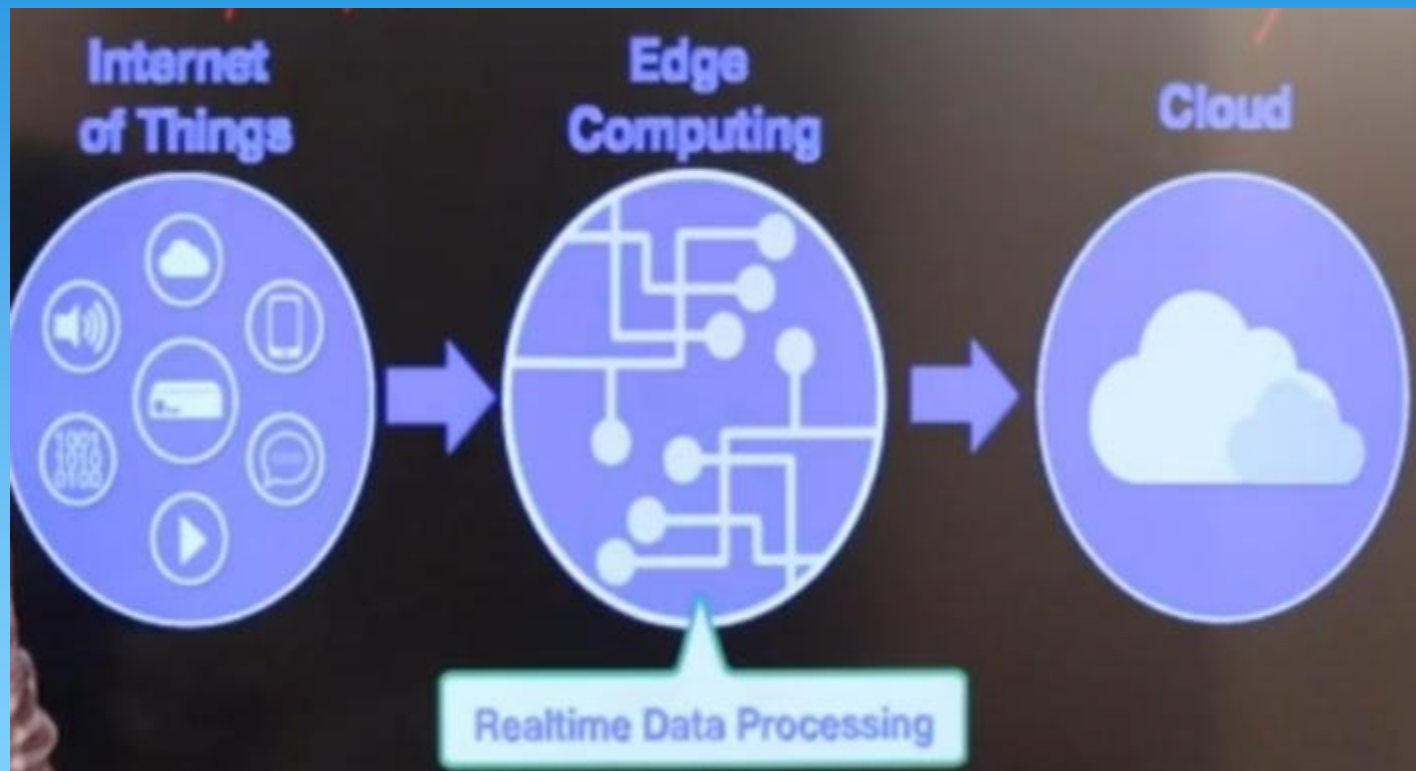


● EDGE COMPUTING :-

- EDGE COMPUTING ENABLES DATA TO BE ANALYZED, PROCESSED & TRANSFERRED AT THE EDGE OF A NETWORK RATHER THAN BEING TRANSMITTED TO A CENTRALISED DATA CENTRE (CLOUD).
- EDGE – WHERE THINGS OR PEOPLE PRODUCE OR CONSUME THAT INFORMATION SUCH AS THE DEVICE ITSELF OR A LOCAL COMPUTER OR SERVER.

Edge Computing





DIFFERENCE BETWEEN EDGE COMPUTING AND CLOUD COMPUTING

1. THE BASIC DIFFERENCE BETWEEN EDGE COMPUTING AND CLOUD COMPUTING LIES IN THE PLACE WHERE THE DATA PROCESSING TAKES PLACE.
2. STORAGE CAPACITY
3. PROCESSING CAPACITY & SPEED
4. LATENCY (LATENCY - THE ABILITY TO PROCESS VERY HIGH VOLUMES OF DATA WITH MINIMAL DELAY)

The cloud

- Big data processing
- Data warehouses



The edge

- Real time data processing
- Local processing



Internet of things

- Smart vehicles
- Connected systems



CHALLENGES OF EDGE COMPUTING

- ESTABLISHING MINI/MICRO DATA CENTERS
- SECURITY OF DEVICES & DATA
- QUALITY & SPEED OF NETWORK (5G)

BENEFITS OF EDGE COMPUTING

- BETTER PERFORMANCE - DECREASEMENT IN LATENCY, IMPROVED CAPACITY
- SECURITY ✓
- DATA EFFICIENT
- LOCALISED SOLUTIONS - MINI/MICRO DATA CENTER
- ENABLES FUTURE TECHNOLOGY - ARTIFICIAL INTELLIGENCE, INTERNET OF THINGS, ROBOTICS

EDGE COMPUTING IS DEVELOPED DUE TO

1. TO ANALYSE DATA LOCALLY, CLOSER TO WHERE IT IS STORED, IN REAL-TIME WITHOUT LATENCY, RATHER THAN SEND IT FAR AWAY TO A CENTRALISED DATA CENTRE.
2. THE EXPONENTIAL GROWTH OF IOT DEVICES, WHICH CONNECT TO THE INTERNET FOR EITHER RECEIVING INFORMATION FROM THE CLOUD OR DELIVERING DATA BACK TO THE CLOUD.
3. MANY IOT DEVICES GENERATE ENORMOUS AMOUNTS OF DATA DURING THE COURSE OF THEIR OPERATION.





